

Testing of hypothesis for sample mean population mean

case-① σ known

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

degree of freedom $(n-1)$

case-②

σ unknown

$n > 30$

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

degree of freedom $(n-1)$

case-③

σ unknown

$n \leq 30$

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

degree of freedom $(n-1)$

$n \rightarrow$ sample size

$s \rightarrow$ sample std. deviation

$\bar{x} \rightarrow$ sample mean

$\sigma \rightarrow$ population std. deviation

$\mu \rightarrow$ population mean

- 10 individuals are chosen @ random from a normal population of students and their marks are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71, 71. In the list of these data discuss the suggestion that mean mark of the population of student is 66. take $\alpha = 0.05$

Ans:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n}$$

H_0 : mean mark is 66, $\mu = 66$

H_1 : mean mark is not 66, $\mu \neq 66$

$\mu = 66$

$$\bar{x} = \frac{\sum x}{n} = \frac{678}{10} = 67.8$$

$$\therefore s^2 = \frac{81.6}{10} = 8.16$$

$$s = 2.85$$

σ is unknown & $n \leq 30$ $\therefore$ case-③

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{67.8 - 66}{2.85/\sqrt{9}} = 1.89$$

from z-table $\Rightarrow$ DOF = $n-1 = 10-1 = 9$

$\alpha = 0.05$

$$\Rightarrow 2.262$$

$$1.89 < 2.262$$

H_0 accepted $\therefore$ mean mark is 66

x	$(x - \bar{x})^2$
63	23.04
63	23.04
66	3.24
67	0.64
68	0.04
69	1.44
70	4.84
70	4.84
71	10.24
71	10.24
$\sum x = 678$	$\sum = 81.6$

- A random sample of 900 members has mean 3.4 cm. Check if it can be reasonably regarded as a sample from a large population of mean 3.2 cm and s.d = 2.3 cm. take $\alpha = 5\%$

Ans:

$$n = 900$$

$$\bar{x} = 3.4$$

$$\mu = 3.2$$

$$\sigma = 2.3$$

$$\sigma \text{ is known } \therefore Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{3.4 - 3.2}{2.3/\sqrt{900}} = \underline{\underline{2.6}}$$

$$H_0: \mu = 3.2 \Rightarrow \text{sample can be regarded}$$

$$H_1: \mu \neq 3.2 \Rightarrow \text{sample cannot be regarded}$$

from table $\Rightarrow$

$$\text{degree of freedom} = n - 1 = 900 - 1 = 899 \quad \&$$

$$\alpha = 0.05$$

$$\Rightarrow \underline{\underline{Z = 1.96}}$$

$$\underline{\underline{2.6}} > 1.96$$

$\therefore H_0$ is rejected.

sample cannot be regarded from that population

- The average height of a particular species of plants is known to be 78 cm. The average ht. of a sample of 100 plants of this species to which particular manure was applied was found to be 80 cm and s.d 10 cm. Examine whether the mean ht. was increased by the application of the particular manure ($\alpha = 0.05$)

Ans:

$$n = 100$$

$$\bar{x} = 80$$

$$s = 10$$

$$\mu = 78$$

$$\alpha = 0.05$$

~~σ~~ or unknown, & ~~all~~ $n > 30$

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{80 - 78}{10/\sqrt{100}} = \underline{\underline{2}}$$

$$H_0: \mu = 78$$

$$H_1: \mu > 78$$

from table $\Rightarrow$

$$\text{degree of freedom} = n - 1 = 100 - 1 = \underline{\underline{99}} \quad \& \quad \alpha = 0.05$$

$$z = \underline{\underline{1.96}} \quad \times$$

$$\underline{\underline{2}} > \underline{\underline{1.96}}$$

$\therefore H_0$ is rejected

$\therefore$ height of the plant is increased by the application of particular manure.

table.	
first row (α)	$H_1 \neq$ ($<$ or $>$) Sample regarded from population difference, same.
second row (α)	$H_1 >$ increase, superior, larger
	$H_1 <$ decrease, inferior, smaller

$$\therefore \text{from table, } z = \underline{\underline{1.645}}$$

$$2 > 1.6$$

$\therefore H_0$ is rejected //

A stenographer claims that she can take dictation @ the rate of 120 words per minute. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words with a std. deviation of 15 words.

Use 5% level of significance.

Ans,

$$n = 100$$

$$\bar{x} = 116$$

$$s = 15$$

$$\alpha = 0.05$$

$$\mu = 120$$

$$H_0: \mu = 120$$

$$H_1: \mu < 120$$

σ unknown, $n > 30 \Rightarrow$ case - (2)

$$\therefore Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{116 - 120}{15/\sqrt{100}} = -2.667$$

from table

$$\text{degree of freedom} = n - 1 = 100 - 1 = 99 \quad \text{error always true} \quad \therefore 2.667 > 1.645$$

$$\Rightarrow \cancel{2.667} \quad Z = 1.645$$

$$\cancel{2.667} > \cancel{1.645} \quad 2.667 > 1.645$$

$\therefore H_0$ is rejected.

~~She can't take dictation @ the rate of 120 words/min~~
Claim is rejected.

~~Test~~
case - (1) σ_1 & σ_2 are known

$$Z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$DF = n_1 + n_2 - 2$$

case - (2) σ_1 & σ_2 unknown; $n_1 + n_2 > 30$

$$Z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$DF = n_1 + n_2 - 2$$

case - (3) σ_1 & σ_2 unknown : $n_1 + n_2 \leq 30$

$$t \text{ or } Z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad \text{DF} = n_1 + n_2 - 2$$

- Test if the mean are significantly different

	size	mean	SD
sample 1	5	11.4	2.65
sample 2	7	14.4	4.37

$$n_1 = 5$$

$$n_2 = 7$$

$$\bar{x}_1 = 11.4$$

$$\bar{x}_2 = 14.4$$

$$H_0 : \bar{x}_1 = \bar{x}_2$$

$$s_1 = 2.65$$

$$s_2 = 4.37$$

$$H_1 : \bar{x}_1 \neq \bar{x}_2$$

$$n_1 + n_2 = 5 + 7 = 12 \leq 30$$

$\therefore$ case - (3)

$$Z = \frac{|11.4 - 14.4|}{\sqrt{\frac{5 \times 2.65^2 + 7 \times 4.37^2}{5 + 7 - 2} \left(\frac{1}{5} + \frac{1}{7} \right)}} = \frac{3}{2.4056}$$
$$= \underline{\underline{1.247}}$$

from table $\Rightarrow$ DF = $n_1 + n_2 - 2 = 5 + 7 - 2 = \underline{\underline{10}}$

$$\alpha = 0.05$$

$$Z = \underline{\underline{2.228}}$$

$$1.247 < 2.228$$

$\therefore H_0$ accepted

means are not significantly different

- The height of 6 randomly chosen sailors are in inches 63, 65, 68, 69, 71 and 72 & those of a randomly chosen soldiers are 61, 62, 65, 66, 69, 70, 71, 72, 73. Test whether sailors are on the avg. taller than soldier.

Ans:

$$H_0: \bar{x}_1 = \bar{x}_2$$

$$H_1: \bar{x}_1 > \bar{x}_2$$

Sailors	
x	$(x - \bar{x})^2$
63	25
65	9
68	0
69	1
71	9
72	16
$\Sigma x = 408$	$\Sigma = 60$

$$\bar{x}_1 = \frac{\Sigma x}{n} = \frac{408}{6} = 68$$

$$n_1 = 6$$

$$s_1^2 = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$= \frac{60}{6} = 10$$

$$s_1 = 3.16$$

Soldiers	
x	$(x - \bar{x})^2$
61	44.49
62	32.14
65	7.12
66	2.79
69	1.77
70	5.42
71	11.09
72	18.74
73	28.40
$\Sigma x = 609$	$\Sigma = 151.96$

$$\bar{x}_2 = \frac{\Sigma x}{n} = \frac{609}{9} = 67.67$$

$$n_2 = 9$$

$$s_2^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{151.96}{9}$$

$$s_2 = 4.1$$

σ_1 & σ_2 unknown : $n_1 + n_2 = 6 + 9 = 15 \leq 30$

$$Z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{|68 - 67.67|}{\sqrt{\frac{6 \times 3.16^2 + 9 \times 4.1^2}{6+9-2} \left(\frac{1}{6} + \frac{1}{9} \right)}}$$

$$= \frac{0.33}{2.1243} = \underline{\underline{0.155}}$$

from table $\Rightarrow$

$$D.F = n_1 + n_2 - 2 = 6 + 9 - 2 = \underline{\underline{13}}$$

$$\alpha = 0.05$$

$$z = \underline{\underline{1.771}}$$

$$0.155 < \underline{\underline{1.771}}$$

H_0 is accepted (both are having same ht.)
 $\therefore$ sailors are not taller than soldier.

- A random sample of size 18 is taken from a normal distribution. Test the hypothesis $H_0: \sigma^2 = 0.36$ against $H_1: \sigma^2 > 0.36$ given $s^2 = 0.68$, $\alpha = 5\%$.

Ans: $H_0: \sigma^2 = 0.36$

$H_1: \sigma^2 > 0.36$

$n = 18$

$s^2 = 0.68$

~~data~~ $n=18$ $s^2=0.68$
 $\sigma^2=0.36$ $\alpha=5\%$

$n \neq \mu \neq n$
 mean $\neq$ variance
 value

$\bar{x}$ & μ is not given $\therefore$ it is variance test

Testing of Variance

Single Variance

χ^2 test or ψ^2 test

$$\chi^2 = \frac{ns^2}{\sigma^2}$$

$$DOF = n - 1$$

Ani: $H_0: \sigma^2 = 0.36$

$$H_1: \sigma^2 > 0.36$$

$$n = 18$$

$$s^2 = 0.68 \quad \sigma^2 = 0.36$$

$$\alpha = 0.05$$

$$\chi^2 = \frac{ns^2}{\sigma^2} = \frac{18 \times 0.68}{0.36} = \underline{\underline{34}}$$

From table $\Rightarrow DOF = n - 1 = 18 - 1 = 17$

$$\alpha = 0.05$$

$$\underline{\underline{\chi^2 = 27.587}}$$

$$\underline{\underline{34 > 27.587}}$$

H_0 rejected

H_1 accepted $\therefore \underline{\underline{\sigma^2 > 0.36}}$

2 independent random samples $n_1 = 10$ & $n_2 = 7$ where observed to have sample variance $s_1^2 = 16$ and $s_2^2 = 3$. Using $\alpha = 0.01$, test whether if there is any significance difference b/w 2 estimates of population variance F-test

Ans:

$$n_1 = 10$$

$$n_2 = 7$$

$$\alpha = 0.01$$

$$s_1^2 = 16$$

$$s_2^2 = 3$$

$$H_0: s_1^2 = s_2^2$$

$$H_1: s_1^2 \neq s_2^2$$

F-test (difference of variance)

$$s_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$$

$$s_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

$$F = \frac{\text{large } s^2}{\text{small } s^2}$$

$$\text{i.e., } F = \frac{s_1^2}{s_2^2} \quad \text{if } s_1^2 > s_2^2$$

for sample data

$$s_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}$$

$$s_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1}$$

$$s_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{10 \times 16}{10 - 1} = \underline{\underline{17.8}}$$

$$s_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{7 \times 3}{7 - 1} = \underline{\underline{3.5}}$$

$$F = \frac{17.8}{3.5} = \underline{\underline{5.08}}$$

$$5.08 < 7.98$$

H_0 accepted

no significant diff. b/w variances

Look $n-1$ value from table $\Rightarrow F = 7.98$

For a random sample of 10 pigs fed on diet A and diet B, the increase in it in a certain period are:

diet A	10	6	16	17	13	12	8	14	15	9		
diet B	7	13	22	15	12	14	18	8	21	23	10	17

S.T the estimates of population variance from sample are not significantly different.

Ans:

x	$\bar{x} = 12$ $(x - \bar{x})^2$	y	$\bar{y} = 15$ $(y - \bar{y})^2$
10	4	7	64
6	36	13	4
16	16	22	49
17	25	15	0
13	1	12	9
12	0	14	1
8	16	18	9
14	4	8	49
15	9	21	36
9	9	23	64
		10	25
		17	4
$\Sigma x = 120$	120	$\Sigma y = 180$	314

$$\bar{x} = \frac{120}{10} = \underline{\underline{12}}$$

$$\bar{y} = \frac{180}{12} = \underline{\underline{15}}$$

$$H_0: s_1^2 = s_2^2$$

$$H_1: s_1^2 \neq s_2^2$$

$$n_1 = 10$$

$$n_2 = 12$$

$$s_1^2 = \frac{\Sigma (x - \bar{x})^2}{n_1 - 1}$$

$$= \frac{120}{9}$$

$$= \underline{\underline{13.33}}$$

$$s_2^2 = \frac{\Sigma (y - \bar{y})^2}{n_2 - 1}$$

$$= \frac{314}{11}$$

$$= \underline{\underline{28.56}}$$

$$F = \frac{28.56}{13.33} = \underline{\underline{2.14}}$$

from table $\Rightarrow \alpha = 0.05$

$$F = \underline{\underline{2.21}}$$

$$2.14 < \underline{\underline{2.21}}$$

H_0 is accepted.

not significantly different

Question paper 2014

- A machine is supposed to produce washers of mean thickness of 0.12 cm. A sample of 10 washers was found to have a mean thickness of 0.12 cm and SD 0.008. Test whether the machine is working in proper order @ 5% of significance level

Ans:

$$n = 10$$

$$\bar{x} = 0.12$$

$$s = 0.008$$

$$\alpha = 0.05$$

$$\mu = 0.12$$

$$H_0: \mu = 0.12$$

$$H_1: \mu \neq 0.12$$

σ unknown $n \leq 30$

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{0.12 - 0.12}{\frac{0.008}{\sqrt{10-1}}} = \underline{\underline{0}}$$

$$DF = n - 1 = 9 ; \alpha = 0.05$$

$$\Rightarrow Z = \underline{\underline{2.262}}$$

$$0 < \underline{\underline{2.622}}$$

H_0 accepted

machine will work properly

5 approximations according to moment in origin table and top values

One variance when samples given

goodness of fit (χ^2 test)

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad DF = n-1$$

- A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class & 20 got a first class. Do these figures commensurate with general examination result which is in the ratio ~~4:3:2:1~~ 4:3:2:1 for various categories respectively

Ans.

$\left. \begin{matrix} \bar{x} \\ H \\ \sigma^2 \\ s^2 \end{matrix} \right\}$ onnum illa $\therefore$ goodness of fit

$$\frac{500}{4+3+2+1} = \frac{500}{10} = 50$$

observed (O)	expected (E)	$\frac{(O-E)^2}{E}$
220	$4 \times 50 = 200$	2
170	$3 \times 50 = 150$	2.667
90	$2 \times 50 = 100$	1
20	$1 \times 50 = 50$	18

23.667

H_0 : compensate

$$\chi^2 = \underline{\underline{23.667}}$$

H_1 : not compensate

from table $\Rightarrow$ DOF = $n-1 = 4-1 = \underline{\underline{3}}$

$$\alpha = 0.05$$

$$\chi^2 = \underline{\underline{7.815}}$$

$$23.667 > 7.815$$

H_0 rejected

observed values not commensurate with general examination results

- the following table gives the no. of accidents that took place in an industry during various days of week. Test if accidents are equally or uniformly distributed over the week

day	mon	tues	wed	thurs	fri	sat
no. of accidents	14	18	12	11	15	14

Ans: Since uniformly distributed,

expected frequency on each day = $14 + 18 + 12 + 11 + 15 + 14$

$\frac{84}{6}$

$= 14$

$$= \frac{84}{6} = 14$$

O	E	$\frac{(O-E)^2}{E}$
14	14	0
18	14	1.142
12	14	0.2857
11	14	0.6428
15	14	0.0714
14	14	0

$$2.1419$$

$$\chi^2 = 2.1419$$

$$DF = 6 - 1 = 5$$

$$\alpha = 0.05$$

$$\chi^2 = 11.0709$$

$$2.1419 < 11.070$$

~~H₀ : accidents~~

H₀ : accidents are uniformly distributed

H₁ : " " " not " "

H₀ accepted

accidents are uniformly distributed //

- On the basis of information given below about the treatment of 200 patients suffering from a disease, state whether new treatment is comparatively superior to the conventional treatment

	favourable	unfavourable	total
new	60	30	90
conventional	40	70	110

Ans

$\frac{90 \times 100}{200} = 45$	$\frac{90 \times 100}{200} = 45$	90
$\frac{110 \times 100}{200} = 55$	$\frac{110 \times 100}{200} = 55$	110
100	100	200

O	E	$\frac{(O-E)^2}{E}$
60	45	5
30	45	5
40	55	4.09
70	55	4.09

$$\chi^2 = 18.18$$

H_0 : new = conventional

H_1 : new > conventional.

~~Ans~~

$$\begin{aligned} \text{D.O.F} &= (n-1)(n-1) && \text{square matrix} \\ &= (2-1)(2-1) && (2 \times 2) \text{ matrix} \\ &= \underline{\underline{1}} \end{aligned}$$

$$\alpha = 0.05$$

$$\chi^2 = \underline{\underline{3.841}}$$

$$18.18 > 3.841$$

H_1 accepted

new is superior than
conventional

- given contingency table for mathematical ability and general ability. find value of χ^2 . Is there good association b/w the 2.

mathematical ability	general ability				
	fair	brown	black	total	
	blue	44	23	5	71
	grey	265	257	178	700
	brown	41	91	98	230
total	350	370	281	1001	

Ans:

	fair	brown	black	total
blue	24.825	26.24	19.93	71
grey	244.76	258.74	196.50	700
brown	80.4	85.01	64.56	230
	350	370	281	1001

O	E	$\frac{(O-E)^2}{E}$
44	24.825	14.81
22	26.24	0.685
5	19.93	11.18
205	244.76	1.673
252	258.74	0.011
178	196.50	1.741
41	80.4	19.30
91	85.01	0.422
98	64.56	17.32

$$\chi^2 = \underline{\underline{67.142}}$$

$$\begin{aligned} DF &= (n-1)(n-1) \\ &= (3-1)(3-1) \\ &= \underline{\underline{4}} \end{aligned}$$

$\Leftarrow (3 \times 3)$ matrix

$$\alpha = 0.05$$

$$\chi^2 = \underline{\underline{9.488}}$$

H_0 : associated

H_1 : not associated

$$67.142 > 9.488$$

H_0 rejected

$\therefore$ both are not associated